

|                                                                                       |                                                    |                        |
|---------------------------------------------------------------------------------------|----------------------------------------------------|------------------------|
| Test Report No.: <b>Xxxxxxxx</b>                                                      |                                                    | Page 1 of xx           |
|                                                                                       |                                                    | Issue Date: DD/MM/YYYY |
| Manufacturer:                                                                         | <b>Applicant Name</b><br>Applicant address         |                        |
| Test item:                                                                            | <b>Wireless Headphone and Earphone</b>             |                        |
| Identification:                                                                       | <b>(Model No.)</b>                                 | Serial No.:            |
| Receipt No.:                                                                          | Date of receipt:                                   | DD/MM/YYYY             |
| Testing laboratory and its address:                                                   | <b>Lab Name</b><br><b>Lab address</b>              |                        |
| Test specification:                                                                   | IS 616:2017/ IEC 60065:2014                        |                        |
| Test Result:                                                                          | The test item Passed/Failed the test specification |                        |
| Other Aspects:                                                                        |                                                    |                        |
| - This report consists of xx pages and attachment as stated in page no. XX            |                                                    |                        |
| This test report relates to the test sample submitted and list of documents attached. |                                                    |                        |

|                             |                                          |                             |
|-----------------------------|------------------------------------------|-----------------------------|
| <b>Tested by:</b>           | <b>Approved by/Authorized Signatory:</b> | <b>Issued by:</b>           |
|                             |                                          |                             |
| <b>(Name / Designation)</b> | <b>(Name / Designation)</b>              | <b>(Name / Designation)</b> |
| <b>Date:</b> DD/MM/YYYY     | <b>Date:</b> DD/MM/YYYY                  | <b>Date:</b> DD/MM/YYYY     |

|                                                                                                                                                                                                               |                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>TEST REPORT</b><br><b>IS 616:2017/ IEC 60065:2014</b><br><b>Audio, Video and Similar Electronic Apparatus - Safety requirements</b><br><b>(Fifth Revision)</b><br><b>“Wireless Headphone and Earphone”</b> |                                      |
| <b>Report Number.</b> .....                                                                                                                                                                                   | XXXXXXX                              |
| <b>Date of issue</b> .....                                                                                                                                                                                    | (See cover page)                     |
| <b>Total number of pages</b> .....                                                                                                                                                                            | (See cover page)                     |
| <b>Testing Laboratory</b> .....                                                                                                                                                                               | Lab Name                             |
| <b>Address</b> .....                                                                                                                                                                                          | Lab Address                          |
| <b>Manufacturer's name</b> .....                                                                                                                                                                              | <b>Applicant's Name</b>              |
| <b>Address</b> .....                                                                                                                                                                                          | Applicant's address                  |
| <b>Test specification:</b>                                                                                                                                                                                    |                                      |
| <b>Standard</b> .....                                                                                                                                                                                         | <b>IS 616: 2017 / IEC 60065:2014</b> |
| <b>Test procedure</b> .....                                                                                                                                                                                   | Compliance Report                    |
| <b>Non-standard test method</b> .....                                                                                                                                                                         | N/A                                  |
| <b>Test Report Form No.</b> .....                                                                                                                                                                             | BIS_AV/WH/WE IS 616_V1.0             |
| <b>Test Report Form(s) Originator</b> ....                                                                                                                                                                    | Bureau of Indian Standards           |
| <b>Test Report Form</b> .....                                                                                                                                                                                 | 01/04/2020                           |
| <b>Test item description</b> ..... <b>Wireless Headphone and Earphone</b>                                                                                                                                     |                                      |
| <b>Trade Mark</b> .....                                                                                                                                                                                       |                                      |
| <b>Model/Type reference</b> .....                                                                                                                                                                             |                                      |
| <b>Ratings</b> .....                                                                                                                                                                                          |                                      |
| <b>Other Documents submitted</b> .....: Please refer to Table – List of Attachments at Page No. xx                                                                                                            |                                      |

|                      |                                   |                      |
|----------------------|-----------------------------------|----------------------|
| Tested by:           | Approved by/Authorized Signatory: | Issued by:           |
|                      |                                   |                      |
| (Name / Designation) | (Name / Designation)              | (Name / Designation) |
| Date: DD/MM/YYYY     | Date: DD/MM/YYYY                  | Date: DD/MM/YYYY     |

**Summary of testing:**

| Test Code | Description            | Measurement / testing                                                                       | Total No. of Tests / Requirement | Total No. of Applicable Tests / Requirement | No. of Tests / Requirement Passed | Page No. |
|-----------|------------------------|---------------------------------------------------------------------------------------------|----------------------------------|---------------------------------------------|-----------------------------------|----------|
| EL 2000   | Marking Requirements   | Marking and Instructions (CI 5)                                                             | 36                               |                                             |                                   |          |
| EL 2001   | Radiation Requirements | Hazardous radiations (CI 6)                                                                 | 06                               |                                             |                                   |          |
| EL 2002   | Heating Requirements   | Heating under normal operating conditions(CI 7)                                             | 08                               |                                             |                                   |          |
| EL 2003   | Electrical safety      | Constructional requirements with regard to the protection against electric shock(CI 8)      | 39                               |                                             |                                   |          |
| EL 2004   | Electrical safety      | Electric shock hazard under normal operating condition (CI 9)                               | 22                               |                                             |                                   |          |
| EL 2005   | Electrical safety      | Insulation requirements (CI 10)                                                             | 08                               |                                             |                                   |          |
| EL 2006   | Electrical safety      | Fault conditions (CI 11)                                                                    | 20                               |                                             |                                   |          |
| EL 2007   | Mechanical properties  | Mechanical Strength (CI 12)                                                                 | 28                               |                                             |                                   |          |
| EL 2008   | Mechanical properties  | Clearances and creepage distances (CI 13)                                                   | 16                               |                                             |                                   |          |
| EL 2009   | Components             | Components (CI 14)                                                                          | 90                               |                                             |                                   |          |
| EL 2010   | Wiring                 | Terminals (CI 15)                                                                           | 34                               |                                             |                                   |          |
| EL 2011   | Wiring                 | External flexible cords (CI 16)                                                             | 14                               |                                             |                                   |          |
| EL 2012   | Wiring                 | Electrical connections and mechanical fixings (CI 17)                                       | 14                               |                                             |                                   |          |
| EL 2013   | Physical properties    | Mechanical strength of picture tubes and protection against the effect of implosion (CI.18) | 03                               |                                             |                                   |          |
| EL 2014   | Physical properties    | Stability and mechanical hazards (CI 19)                                                    | 14                               |                                             |                                   |          |

|         |                                    |                                                                              |    |  |  |  |
|---------|------------------------------------|------------------------------------------------------------------------------|----|--|--|--|
| EL 2015 | Physical properties                | Resistance to fire (CI 20)                                                   | 16 |  |  |  |
| EL 2016 | Protection against splashing water | Requirements for apparatus with protection against splashing water (Annex A) | 07 |  |  |  |
| EL 2017 | Communicating connection           | Apparatus to be connected to the TELECOMMUNICATION NETWORKS (Annex B)        | 09 |  |  |  |
| EL 2018 | Insulation Properties              | Measuring network for TOUCH CURRENTS (Annex D)                               | 02 |  |  |  |
| EL 2019 | Electrical Safety                  | Measurement of Clearances and Creepage Distances(Annex E)                    | 01 |  |  |  |
| EL 2020 | Electrical Safety                  | Table of electrochemical potentials (Annex F)                                | 01 |  |  |  |
| EL 2021 | Electrical Safety                  | Flammability test methods (Annex G)                                          | 08 |  |  |  |
| EL 2022 | Insulation Properties              | Insulated Winding Wires For Use Without Interleaved Insulation (Annex H)     | 10 |  |  |  |
| EL 2023 | Electrical safety                  | Alternative Method For Determining Minimum Clearances(Annex J)               | 10 |  |  |  |
| EL 2024 | Electrical safety                  | Impulse Test Generators (Annex K)                                            | 02 |  |  |  |
| EL 2025 | Photographic Purposes              | Electronic Flash Apparatus For Photographic Purposes (Annex L)               | 10 |  |  |  |

**Certificate:** It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

.....  
(Approving Authority)

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**Table EL 2009-(01 to 83)**

| <b>14</b>                  | <b>TABLE: Critical components information</b> |                     |                       |                 |                              |
|----------------------------|-----------------------------------------------|---------------------|-----------------------|-----------------|------------------------------|
| <b>Object / part No.</b>   | <b>Manufacturer/ trademark</b>                | <b>Type / model</b> | <b>Technical data</b> | <b>Standard</b> | <b>Mark(s) of conformity</b> |
|                            |                                               |                     |                       |                 |                              |
| <b>- Description:</b>      |                                               |                     |                       |                 |                              |
|                            |                                               |                     |                       |                 |                              |
|                            |                                               |                     |                       |                 |                              |
|                            |                                               |                     |                       |                 |                              |
| <b>- Description:</b>      |                                               |                     |                       |                 |                              |
|                            |                                               |                     |                       |                 |                              |
|                            |                                               |                     |                       |                 |                              |
|                            |                                               |                     |                       |                 |                              |
| <b>- Description:</b>      |                                               |                     |                       |                 |                              |
|                            |                                               |                     |                       |                 |                              |
|                            |                                               |                     |                       |                 |                              |
| Supplementary information: |                                               |                     |                       |                 |                              |

Tested by:

Approved by/ Authorized Signatory:

Name/Designation

Name/Designation

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**Test item particulars.....:****Sample received condition.....:** ☐ Good ☐ Others**Classification of installation and use..... :** Class I / Class II**Supply Connection .....** :

..... :

**Laboratory conditions.....:****Ambient Temperature .....** : 25±5°C**Ambient Humidity .....** : 40 to 75 %**Testing:****Date of receipt of test item .....** :**Date (s) of performance of tests .....** :**Table – List of Attachments**

| Attachment No. | Attachment Description | No. of pages in Attachment |
|----------------|------------------------|----------------------------|
| Attachment – 1 | Photo Documentation    | xx                         |

**Possible test case verdicts:**

- test case does not apply to the test object.....: N/A
- test object does meet the requirement.....: P (Pass)
- test object does not meet the requirement .....: F (Fail)

**General remarks:**

The test results presented in this report relate only to the object tested.

This report shall not be reproduced, except in full, without the written approval of the Issuing testing laboratory.

**General product information:****Differences between the models:****Model No. tested with-in the family series** :

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| 3   | General requirements                                                                                                                                                                 | Verdict |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|     | Safety class of the apparatus ..... :                                                                                                                                                |         |
| 3.3 | Constructions and components not specifically covered                                                                                                                                |         |
|     | Equipment involving technologies, components and materials or methods of construction not specifically covered in this standard, comply with the safety requirement of this standard |         |
| 3.4 | Components and subassemblies that comply with IEC 62368-1                                                                                                                            |         |

| 4         | General test conditions                                                                                                                                                                                                                                                                                                                                                                                                                      | Verdict |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 4.1.4     | Ventilation instructions require the use of the test box                                                                                                                                                                                                                                                                                                                                                                                     |         |
| 4.2.2     | Measurement of rated current consumption or rated power consumption of television sets as specified.                                                                                                                                                                                                                                                                                                                                         |         |
| 4.2.5 (e) | Equipment containing multi-channel amplifiers, where each channel can be operated independently using rated load impedance at output power level that corresponds to 1/8 <sup>th</sup> of non-clipped output power and the channel that cannot be operated independently shall be operated using rated load impedance at output power level in such a way to deliver one-eighth of the non-clipped output power to the rated load impedance. |         |

|                  |                                    |
|------------------|------------------------------------|
| Tested by:       | Approved by/ Authorized Signatory: |
| Name/Designation | Name/Designation                   |

**Tests relating to Marking Requirements****EL 2000-V1.0**

| Clause No. | Test / Requirement name                                                                                                                                                                                                                                                                 | Test Code  | Test result/ observation | Verdict |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------|---------|
| 5          | Marking and instructions*                                                                                                                                                                                                                                                               | EL 2000-00 |                          |         |
| 5.1        | General requirements                                                                                                                                                                                                                                                                    | EL 2000-01 |                          |         |
|            | Comprehensible and easily discernible*                                                                                                                                                                                                                                                  | EL 2000-02 |                          |         |
|            | Permanent durability against water and petroleum spirit                                                                                                                                                                                                                                 | EL 2000-03 |                          |         |
| 5.2        | Identification and supply ratings*                                                                                                                                                                                                                                                      |            |                          |         |
|            | a) Identification, maker *                                                                                                                                                                                                                                                              | EL 2000-04 |                          |         |
|            | b) Model number or type reference*                                                                                                                                                                                                                                                      | EL 2000-05 |                          |         |
|            | c) Class II symbol or Class II with functional earth symbol if applicable *                                                                                                                                                                                                             | EL 2000-06 |                          |         |
|            | d) Nature of supply*                                                                                                                                                                                                                                                                    | EL 2000-07 |                          |         |
|            | e) Rated supply voltage*                                                                                                                                                                                                                                                                | EL 2000-08 |                          |         |
|            | f) Mains frequency if safety dependant *                                                                                                                                                                                                                                                | EL 2000-09 |                          |         |
|            | g) Rated current or power consumption for apparatus supplied by supply apparatus for general use, on apparatus or in instruction manual:<br>Measured current or power consumption:<br>Deviation % (max 10%):*                                                                           | EL 2000-10 |                          |         |
|            | h) Rated current or power consumption for apparatus intended for connection to an a.c. mains supply:<br>Measured current or power consumption for Television set:<br>Deviation % (max 10%): *                                                                                           | EL 2000-11 |                          |         |
|            | appliance coupler for Class I is used for Class II equipment with functional earth connection, the requirements of Clause 15 and Clause 16 related to Class I construction shall be applied up to the connecting point of the protective (earthing) conductor to the functional earth.* | EL 2000-12 |                          |         |

**Tests relating to Marking Requirements****EL 2000-V1.0**

| Clause No. | Test / Requirement name                                                                                                                                                                                                                                          | Test Code  | Test result/ observation | Verdict |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------|---------|
|            | Graphical symbols placed on the apparatus, whether required by this standard or not, shall be in accordance with IEC 60417 or ISO 3864-2 or ISO 7000, if available. In the absence of suitable symbols, the manufacturer may design specific graphical symbols.* | EL 2000-13 |                          |         |
|            | Care shall be taken so that additional markings and instructions not required by this standard do not contradict the markings and instructions required by this standard. Symbols placed on the equipment shall be explained in the user manual.*                | EL 2000-14 |                          |         |
| 5.3        | Terminals*                                                                                                                                                                                                                                                       |            |                          |         |
|            | a) Earth terminal*                                                                                                                                                                                                                                               | EL 2000-15 |                          |         |
|            | b) Hazardous live terminals*                                                                                                                                                                                                                                     | EL 2000-16 |                          |         |
|            | c) Markings on supply output terminals*                                                                                                                                                                                                                          | EL 2000-17 |                          |         |
| 5.4        | Caution marking*                                                                                                                                                                                                                                                 |            |                          |         |
|            | a) Use of triangle with exclamation mark*                                                                                                                                                                                                                        | EL 2000-18 |                          |         |
|            | b) Marking on loudspeaker grille, IEC 60417-5036*                                                                                                                                                                                                                | EL 2000-19 |                          |         |
|            | c) User-replaceable coin / button cell battery marking *                                                                                                                                                                                                         | EL 2000-20 |                          |         |
| 5.5        | Instructions*                                                                                                                                                                                                                                                    | EL 2000-21 |                          |         |
| 5.5.1      | Safety relevant information*                                                                                                                                                                                                                                     | EL 2000-22 |                          |         |
| 5.5.2      | a) Mains powered equipment not exposed to dripping or splashing. Warning concerning objects filled with liquid, etc. *                                                                                                                                           | EL 2000-23 |                          |         |
|            | b) Hazardous live terminals, instructions for wiring *                                                                                                                                                                                                           | EL 2000-24 |                          |         |
|            | c) Instructions for replacing lithium battery *                                                                                                                                                                                                                  | EL 2000-25 |                          |         |
|            | d) Class I earth connection warning *                                                                                                                                                                                                                            | EL 2000-26 |                          |         |

**Tests relating to Marking Requirements****EL 2000-V1.0**

| Clause No. | Test / Requirement name                                                                              | Test Code  | Test result/ observation | Verdict |
|------------|------------------------------------------------------------------------------------------------------|------------|--------------------------|---------|
|            | e) Instructions for multimedia system connection *                                                   | EL 2000-27 |                          |         |
|            | f) Special stability warning for attachment of the apparatus to the floor/wall*                      | EL 2000-28 |                          |         |
|            | g) ) Warning: battery exposure to heat *                                                             | EL 2000-29 |                          |         |
|            | h) Warning: protective film on CRT face *                                                            | EL 2000-30 |                          |         |
|            | i) Warning: Non-floor standing TV >7kg*                                                              | EL 2000-31 |                          |         |
|            | j) Warning: User replaceable coin / button cell battery*                                             | EL 2000-32 |                          |         |
| 5.5.3      | a-b) Disconnect device: plug/coupler or all-pole mains switch location, accessibility and markings * | EL 2000-33 |                          |         |
|            | c) Instructions for permanently connected equipment*                                                 | EL 2000-34 |                          |         |
|            | Marking, signal lamps or similar for completely disconnection from the mains *                       | EL 2000-35 |                          |         |

\* Total number of Requirements to be observed / inspected= 34

Total No. of Applicable Requirement =

No of Requirements for which the sample passed =

Total number of tests to be conducted = 02

Total No. of Applicable Tests =

No. of tests for which the sample passed =

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

.....  
(Approving Authority)

**Tests relating to Radiation Requirements****EL 2001-V1.0**

| Clause No | Test / Requirement name                                                      | Code       | Test result/ observation | Verdict |
|-----------|------------------------------------------------------------------------------|------------|--------------------------|---------|
| 6         | Hazardous radiation*                                                         | EL 2001-00 |                          |         |
| 6.1       | Ionizing radiation < 36 pA/kg<br>(0,5 mR/h)                                  |            |                          |         |
|           | Ionizing radiation under fault condition                                     | EL 2001-01 |                          |         |
| 6.2       | Laser radiation, emission limits to<br>IEC 60825-1:2007                      | EL 2001-02 |                          |         |
|           | Emission limits under normal operating<br>conditions as per IEC 60825-1:2007 | EL 2001-03 |                          |         |
|           | Emission limits under fault<br>conditions as per IEC 60825-1                 | EL 2001-04 |                          |         |
| 6.3       | Light emitting diodes (LEDs) according<br>to IEC 62471                       | EL 2001-05 |                          |         |

\*Total number of Requirements to be observed / inspected = 01

Total No. of Applicable Requirement =

No of Requirements for which the sample passed =

Total number of tests to be conducted = 05

Total No. of Applicable Tests =

No. of tests for which the sample passed =

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

.....  
(Approving Authority)

**Tests relating to Heating Requirements****EL 2002-V1.0**

| Clause No | Test / Requirement name                                                                                                                               | Code       | Test result/ observation | Verdict |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------|---------|
| 7         | Heating under normal operating conditions*                                                                                                            | EL 2002-00 |                          |         |
| 7.1       | General                                                                                                                                               |            |                          |         |
| 7.1.1     | Temperature rises not exceeding specified values; fuse links and other protective devices defeated                                                    | EL 2002-01 |                          |         |
| 7.1.2     | Temperature rise of accessible parts                                                                                                                  | EL 2002-02 |                          |         |
| 7.1.3     | Temperature rise of parts providing electrical insulation                                                                                             | EL 2002-03 |                          |         |
| 7.1.4     | Temperature rise of parts acting as a support or as a mechanical barrier                                                                              | EL 2002-04 |                          |         |
| 7.1.5     | Temperature rise of windings                                                                                                                          | EL 2002-05 |                          |         |
| 7.1.6     | Parts not subject to a limit under 7.1.1 to 7.1.4                                                                                                     | EL 2002-06 |                          |         |
| 7.2       | Softening temperature of insulating material supporting parts conductively connected to the mains carrying a current > 0.2 A, shall be at least 150°C | EL 2002-07 |                          |         |

\*Total number of Requirements to be observed / inspected = 01

Total No. of Applicable Requirement =

No of Requirements for which the sample passed =

Total number of tests to be conducted = 07

Total No. of Applicable Tests =

No. of tests for which the sample passed =

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

.....  
(Approving Authority)

**Tests relating to Electrical Safety****EL 2003- V1.0**

| Clause No | Test / Requirement name                                                                                                                         | Code       | Test result/ observation | Verdict |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------|---------|
| 8         | Constructional requirements with regard to the protection against electric shock*                                                               | EL 2003-00 |                          |         |
| 8.1       | Conductive parts covered by lacquer, paper, untreated textile oxide films and beads etc. considered to be bare *                                | EL 2003-01 |                          |         |
| 8.2       | No shock hazard when changing voltage setting device, fuse-links or handling drawers etc.                                                       | EL 2003-02 |                          |         |
| 8.3       | Insulation of hazardous live parts not provided by hygroscopic material                                                                         | EL 2003-03 |                          |         |
| 8.4       | No risk of electric shock from accessible parts or from parts rendered accessible following the removal of a cover which can be removed by hand | EL 2003-04 |                          |         |
| 8.5       | Class I apparatus*                                                                                                                              |            |                          |         |
|           | Basic insulation between hazardous live parts and earthed accessible parts *                                                                    | EL 2003-05 |                          |         |
|           | Resistors bridging basic insulation shall complying with 14.1 a) *                                                                              | EL 2003-06 |                          |         |
|           | Capacitors bridging basic insulation shall complying with 14.2.1 a) *                                                                           | EL 2003-07 |                          |         |
|           | Protective earthing terminal *                                                                                                                  | EL 2003-08 |                          |         |
| 8.6       | Class II apparatus                                                                                                                              |            |                          |         |
|           | a) Basic and supplementary insulation between hazardous live parts and accessible parts *                                                       | EL 2003-09 |                          |         |
|           | b) Reinforced insulation between hazardous live parts and accessible parts *                                                                    | EL 2003-10 |                          |         |
| 8.7       | Components bridging insulation                                                                                                                  |            |                          |         |
|           | Basic insulation bridged by components complying with 14.4.5.3                                                                                  | EL 2003-11 |                          |         |
|           | Components bridging basic, supplementary, double or reinforced insulation complying with 14.2 a) or 14.4                                        | EL 2003-12 |                          |         |
|           | Basic and supplementary insulation each being bridged by a capacitor or RC-unit complying with 14.3.2 a)                                        | EL 2003-13 |                          |         |
|           | Double or reinforced insulation being bridged with 2 capacitors or RC-units in series complying with 14.3.2 a)                                  | EL 2003-14 |                          |         |

|      |                                                                                                                                                                            |            |  |  |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--|--|
|      | Double or reinforced insulation being bridged with a single capacitor or RC-unit complying with 14.3.2 b)                                                                  | EL 2003-15 |  |  |
| 8.8  | Insulation thickness and thin sheet materials                                                                                                                              |            |  |  |
|      | Basic or supplementary insulation > 0,4 mm (mm)*                                                                                                                           | EL 2003-16 |  |  |
|      | Reinforced insulation > 0,4 mm (mm) *                                                                                                                                      | EL 2003-17 |  |  |
|      | Thin sheet material used inside the equipment                                                                                                                              | EL 2003-18 |  |  |
|      | Basic or supplementary insulation, at least two layers, each meeting 10.4                                                                                                  | EL 2003-19 |  |  |
|      | Basic or supplementary insulation, three layers any two of which meet 10.4                                                                                                 | EL 2003-20 |  |  |
|      | Reinforced insulation, two layers each of which meet 10.4                                                                                                                  | EL 2003-21 |  |  |
| 8.9  | Adequate insulation between internal hazardous live conductors and accessible parts, or between internal hazardous live parts and conductors connected to accessible parts | EL 2003-22 |  |  |
| 8.10 | Double insulation between accessible parts and conductors connected to the mains                                                                                           | EL 2003-23 |  |  |
|      | Double insulation between conductors connected to accessible parts and parts connected to the mains                                                                        | EL 2003-24 |  |  |
| 8.11 | Detaching of wires                                                                                                                                                         |            |  |  |
|      | No undue reduction of creepage or clearance distances if wires become detached                                                                                             | EL 2003-25 |  |  |
|      | Vibration test carried out                                                                                                                                                 | EL 2003-26 |  |  |
| 8.12 | Adequate fastening of windows, lenses, lamp covers etc. (pull test 20 N for 10 s)                                                                                          | EL 2003-27 |  |  |
| 8.13 | Adequate fastening of windows, lenses, lamp covers etc. (pull test 20 N for 10 s)                                                                                          | EL 2003-28 |  |  |
| 8.14 | No risk of damage to the insulation of internal wiring due to hot parts or sharp edges                                                                                     | EL 2003-29 |  |  |
| 8.15 | Only special supply equipment can be used*                                                                                                                                 | EL 2003-30 |  |  |
| 8.16 | Insulated winding wire without additional interleaved insulation                                                                                                           | EL 2003-31 |  |  |
| 8.17 | Endurance test as required by 8.16                                                                                                                                         | EL 2003-32 |  |  |
| 8.18 | Disconnection from the mains                                                                                                                                               |            |  |  |
|      | Disconnect device used in apparatus                                                                                                                                        | EL 2003-33 |  |  |
|      | All-pole switch or circuit breaker with >3mm contact separation.                                                                                                           | EL 2003-34 |  |  |
|      | Mains switch ON indication *                                                                                                                                               | EL 2003-35 |  |  |
| 8.19 | Switch not fitted in the mains cord *                                                                                                                                      | EL 2003-36 |  |  |

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|      |                                           |            |  |  |
|------|-------------------------------------------|------------|--|--|
| 8.20 | Bridging components comply with clause 14 | EL 2003-37 |  |  |
| 8.21 | Non-separable thin sheet material         | EL 2003-38 |  |  |

\*Total number of Requirements to be observed / inspected = 13  
Total No. of Applicable Requirement =  
No of Requirements for which the sample passed =

Total number of tests to be conducted = 26  
Total No. of Applicable Tests =  
No. of tests for which the sample passed =

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

.....  
(Approving Authority)

**Tests relating to Electrical Safety****EL 2004-V1.0**

| Clause No | Test / Requirement name                                                                   | Code       | Test result/ observation | Verdict |
|-----------|-------------------------------------------------------------------------------------------|------------|--------------------------|---------|
| 9         | Electric shock hazard under normal operating conditions*                                  | EL 2004-00 |                          |         |
| 9.1       | Testing on the outside                                                                    | EL 2004-01 |                          |         |
| 9.1.1     | Requirements                                                                              |            |                          |         |
|           | Accessible parts shall not be hazardous live                                              | EL 2004-02 |                          |         |
|           | Inaccessible terminals are not accessible or comply with relevant requirements            | EL 2004-03 |                          |         |
|           | For voltages >1000 V ac or >1500 V dc complies with clause 13.3.1 for basic insulation    | EL 2004-04 |                          |         |
| 9.1.1.2   | Determination of hazardous live parts                                                     |            |                          |         |
|           | a) Open circuit voltages                                                                  | EL 2004-05 |                          |         |
|           | b) Touch current measured from terminal devices using the network in annex D              | EL 2004-06 |                          |         |
|           | c) The charge exceeds 45 $\mu\text{C}$                                                    | EL 2004-07 |                          |         |
|           | d) Energy of discharge not exceeding 350 mJ                                               | EL 2004-08 |                          |         |
| 9.1.1.3   | Test with test finger and test probe                                                      | EL 2004-09 |                          |         |
| 9.1.2     | No hazardous live shafts of knobs, handles or levers                                      | EL 2004-10 |                          |         |
| 9.1.3     | Ventilation holes and other holes tested by means of 4 mm x 100 mm test pin               | EL 2004-11 |                          |         |
| 9.1.4     | Terminal devices tested with 1 mm x 20 mm test pin (10 N); test probe D of IEC 61032      | EL 2004-12 |                          |         |
|           | Terminal devices tested with 1 mm x 100 mm straight wire (1 N); test probe D of IEC 61032 | EL 2004-13 |                          |         |
| 9.1.5     | Pre-set controls tested with 2.5 mm x 100 mm test pin (10 N); test probe C of IEC 61032   | EL 2004-14 |                          |         |
| 9.1.6     | Withdrawal of the mains plug:                                                             |            |                          |         |
|           | No shock hazard due to stored charge after 2 s :                                          | EL 2004-15 |                          |         |
|           | Bleeder resistor(s) comply with 14.2 or no shock hazard when open circuited               | EL 2004-16 |                          |         |
|           | If C is not greater than 0,1 $\mu\text{F}$ no test needed                                 | EL 2004-17 |                          |         |
| 9.1.7     | Resistance to external forces                                                             |            |                          |         |

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|     |                                                    |            |  |  |
|-----|----------------------------------------------------|------------|--|--|
|     | a) Test probe 11 of IEC 61032 for 10 s (50 N)      | EL 2004-18 |  |  |
|     | b) Test hook of fig. 4 for 10 s (20 N)             | EL 2004-19 |  |  |
|     | c) 30 mm diameter test tool for 5 s (100 or 250 N) | EL 2004-20 |  |  |
| 9.2 | No hazard after removing a cover by hand           | EL 2004-21 |  |  |

Total number of requirements to be observed / inspected = 01

Total No. of Applicable Requirement =

No of requirements for which sample passed =

Total number of tests to be conducted = 21

Total No. of Applicable Tests =

No. of tests for which the sample passed =

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

.....  
(Approving Authority)

**Tests relating to Electrical Safety****EL 2005- V1.0**

| Clause No | Test / Requirement name                                                                                                                                                                                                                                       | Code       | Test result/ observation | Verdict |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------|---------|
| 10        | Insulation requirements*                                                                                                                                                                                                                                      | EL 2005-00 |                          |         |
| 10.2      | Surge Test                                                                                                                                                                                                                                                    |            |                          |         |
|           | a) Insulation between desired terminals is subjected to 50 discharges at a maximum rate of 12/min, from 1 nF capacitor charged to 10 kV in a test circuit as per Fig.5a.                                                                                      | EL 2005-01 |                          |         |
|           | b) After surge test, the tested insulation shall comply with clause 10.3, Table 5 :<br>i) Insulation resistance : 2 MΩ, Min. (for basic insulation) ,<br>ii) Insulation resistance :4 MΩ, Min. (for reinforced insulation) ,<br>iii) Dielectric strength test | EL 2005-02 |                          |         |
| 10.3      | Humidity treatment 48 h or 120 h                                                                                                                                                                                                                              | EL 2005-03 |                          |         |
| 10.4      | Insulation resistance and dielectric strength                                                                                                                                                                                                                 |            |                          |         |
| 10.4.1    | Insulation of the insulating materials                                                                                                                                                                                                                        | EL 2005-04 |                          |         |
| 10.4.2    | Between parts of different polarity directly connected to the mains                                                                                                                                                                                           | EL 2005-05 |                          |         |
|           | Between parts separated by BASIC or SUPPLEMENTARY insulation                                                                                                                                                                                                  | EL 2005-06 |                          |         |
|           | Between parts separated by REINFORCED insulation                                                                                                                                                                                                              | EL 2005-07 |                          |         |

Total number of requirement to be observed / inspected = 01

Total No. of Applicable Requirement =

No of requirement for which the sample passed =

Total number of tests to be conducted = 07

Total No. of Applicable Tests =

No. of tests for which the sample passed =

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

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(Approving Authority)

**Tests relating to Electrical Safety****EL 2006- V1.0**

| Clause No | Test / Requirement name                                                                                                                                          | Code       | Test result/ observation | Verdict |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------|---------|
| 11        | Fault conditions*                                                                                                                                                | EL 2006-00 |                          |         |
| 11.1      | No shock hazard under fault condition                                                                                                                            |            |                          |         |
|           | a) - The permissible values of Open circuit voltage [as per Cl.9.1.1.1(a)] for other than audio signals, are increased to 70 V(peak) a.c. and 120 V d.c.         | EL 2006-01 |                          |         |
|           | b) - The permissible values of Touch Current [Cl.9.1.1.1(b)] are increased to $U_1=70V$ (peak) and $U_2=1.4V$ (peak) for a.c. and to $U_1=4 V$ for d.c.          | EL 2006-02 |                          |         |
| 11.2      | Heating                                                                                                                                                          | EL 2006-03 |                          |         |
| 11.2.1    | Requirements                                                                                                                                                     |            |                          |         |
|           | No danger of fire to the surroundings                                                                                                                            | EL 2006-04 |                          |         |
|           | Safety not impaired by abnormal heat                                                                                                                             | EL 2006-05 |                          |         |
|           | Flames extinguish within 10 seconds                                                                                                                              | EL 2006-06 |                          |         |
|           | No hazard from softening solder                                                                                                                                  | EL 2006-07 |                          |         |
|           | Soldered terminations not used as protective mechanism                                                                                                           | EL 2006-08 |                          |         |
| 11.2.2    | Measurement of temperature rises                                                                                                                                 | EL 2006-09 |                          |         |
| 11.2.3    | Temperature rise of accessible parts                                                                                                                             | EL 2006-10 |                          |         |
| 11.2.4    | Temperature rise of parts, other than windings and printed boards, providing electrical insulation                                                               | EL 2006-11 |                          |         |
| 11.2.5    | Temperature rise of parts acting as a support or mechanical barrier                                                                                              | EL 2006-12 |                          |         |
| 11.2.6    | Temperature rise of windings                                                                                                                                     | EL 2006-13 |                          |         |
| 11.2.7    | Printed boards                                                                                                                                                   |            |                          |         |
|           | Temperature rise does not exceed the limits of table 3 or exceed the limits of table 3 by max. 100 K for max. 5 min                                              | EL 2006-14 |                          |         |
|           | a) Temperature rise of V-0 or VTM-0 printed circuit boards exceeding the limits of table 3 by not more than 100 K for an area not greater than 2 cm <sup>2</sup> | EL 2006-15 |                          |         |

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|        |                                                                                                                                                                              |            |  |  |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--|--|
|        | b) Temperature rise of V-0 or VTM-0 printed circuit boards exceeding the limits of table 3 up to 300 K for an area not greater than 2 cm <sup>2</sup> for a maximum of 5 min | EL 2006-16 |  |  |
|        | Meets all the special conditions if conductors on printed circuit boards are interrupted                                                                                     | EL 2006-17 |  |  |
|        | Class I protective earthing maintained                                                                                                                                       | EL 2006-18 |  |  |
| 11.2.8 | Temperature rise of parts not subject to the limits of 11.2.2 to 11.2.7 shall not exceed the limits in table 3, item e), "Fault conditions".                                 | EL 2006-19 |  |  |

Total number of requirements to be observed / inspected = 01

Total No. of Applicable Requirement =

No of requirements for which the sample passed =

Total number of tests to be conducted = 19

Total No. of Applicable Tests =

No. of tests for which the sample passed =

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

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(Approving Authority)

**Tests relating to Mechanical Properties****EL 2007- V1.0**

| Clause No | Test / Requirement name                                                                         | Code       | Test result/ observation | Verdict |
|-----------|-------------------------------------------------------------------------------------------------|------------|--------------------------|---------|
| 12        | Mechanical strength*                                                                            | EL 2007-00 |                          |         |
| 12.1.1    | The apparatus have adequate mechanical strength                                                 |            |                          |         |
| 12.1.2    | Bump test where mass >7 kg                                                                      | EL 2007-01 |                          |         |
| 12.1.3    | Vibration test                                                                                  | EL 2007-02 |                          |         |
| 12.1.4    | Impact test                                                                                     | EL 2007-03 |                          |         |
|           | Impact hammer test                                                                              | EL 2007-04 |                          |         |
|           | Steel ball test                                                                                 | EL 2007-05 |                          |         |
| 12.1.5    | Drop test for portable appliance where mass ≤7 kg                                               | EL 2007-06 |                          |         |
| 12.1.6    | Thermoplastic enclosures stress relief test                                                     | EL 2007-07 |                          |         |
| 12.2      | Fixing of knobs, push buttons, keys and levers                                                  | EL 2007-08 |                          |         |
| 12.3      | Remote control hazardous live parts                                                             | EL 2007-09 |                          |         |
| 12.4      | Drawers ( pull test 50N,10s)                                                                    | EL 2007-10 |                          |         |
| 12.5      | Antenna coaxial sockets and providing isolation                                                 | EL 2007-11 |                          |         |
|           | a) - Endurance test,                                                                            | EL 2007-12 |                          |         |
|           | b) - Impact test,                                                                               | EL 2007-13 |                          |         |
|           | c) - Torque test                                                                                | EL 2007-14 |                          |         |
| 12.6      | Telescoping or rod antennas                                                                     | EL 2007-15 |                          |         |
| 12.6.1    | 6,0mm diameter end                                                                              | EL 2007-16 |                          |         |
|           | Prevented from falling into the apparatus                                                       | EL 2007-17 |                          |         |
| 12.6.2    | Physical securement, removal prevented                                                          | EL 2007-18 |                          |         |
| 12.7      | Apparatus containing coin / button cell batteries                                               | EL 2007-19 |                          |         |
| 12.7.1    | Coin/button cell batteries with a diameter of 32 mm or less likely to be ACCESSIBLE to children | EL 2007-20 |                          |         |
| 12.7.2    | Reduced possibility for children to remove battery                                              | EL 2007-21 |                          |         |
| 12.7.3    | Tests                                                                                           |            |                          |         |
| 12.7.3.2  | Stress relief test                                                                              | EL 2007-22 |                          |         |
| 12.7.3.3  | Battery replacement test                                                                        | EL 2007-23 |                          |         |
| 12.7.3.4  | Drop test                                                                                       | EL 2007-24 |                          |         |
| 12.7.3.5  | Impact test                                                                                     | EL 2007-25 |                          |         |

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|          |                                                            |            |  |  |
|----------|------------------------------------------------------------|------------|--|--|
| 12.7.3.6 | Crush test                                                 | EL 2007-26 |  |  |
| 12.7.4   | The battery compartment door/cover shall remain functional | EL 2007-27 |  |  |

Total number of requirements to be observed / inspected = 01

Total No. of Applicable Requirement =

No of requirements for which the sample passed =

Total number of tests to be conducted = 27

Total No. of Applicable Tests =

No. of tests for which the sample passed =

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

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(Approving Authority)

**Tests relating to Mechanical Properties****EL 2008- V1.0**

| Clause No | Test / Requirement name                                                                                                                                                                                                                                             | Code       | Test result/ observation | Verdict |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------|---------|
| 13        | Clearances and creepage distances*                                                                                                                                                                                                                                  | EL 2008-00 |                          |         |
| 13.1      | General                                                                                                                                                                                                                                                             |            |                          |         |
| 13.2      | Determination of Working voltage                                                                                                                                                                                                                                    | EL 2008-01 |                          |         |
| 13.3.2    | Circuits conductively connected to the mains comply with table 8 and, where applicable, table 9:                                                                                                                                                                    | EL 2008-02 |                          |         |
| 13.3      | Clearances                                                                                                                                                                                                                                                          |            |                          |         |
| 13.3.1    | Comply with 13.3 or Annex J                                                                                                                                                                                                                                         |            |                          |         |
| 13.3.3    | Circuits not conductively connected to the mains comply with table 10                                                                                                                                                                                               | EL 2008-03 |                          |         |
| 13.3.4    | Measurement of transient voltages                                                                                                                                                                                                                                   | EL 2008-04 |                          |         |
| 13.4      | a) Creepage distances shall be not less than the appropriate minimum values specified in Table 11, taking into account the values of operating voltage, the pollution degree and the material group.                                                                | EL 2008-05 |                          |         |
|           | When determining the operating voltage for a TNV circuit connected to a telecommunication network whose characteristics are not known, the normal operating voltages shall be assumed to be as : 60 V dc for TNV-1 circuits,<br>120 V dc for TNV-2 & TNV-3 circuits | EL 2008-06 |                          |         |
|           | c) - Classification of Material groups :<br>Material group I – $600 \leq CTI$<br>Material group II - $400 \leq CTI < 600$<br>Material group IIIa - $175 \leq CTI < 400$<br>Material group IIIb - $100 \leq CTI < 175$                                               | EL 2008-07 |                          |         |
| 13.5.1    | Clearances and creepage distances between conductors on printed circuit boards, one of which may be conductively connected to the mains, as in fig. 10                                                                                                              | EL 2008-08 |                          |         |
| 13.5.2    | Type B coated printed circuit boards complying with IEC 60664-3 (basic insulation only)                                                                                                                                                                             | EL 2008-09 |                          |         |

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|      |                                                                                                                                                                                                                                             |            |  |  |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--|--|
| 13.6 | a) - Conductive parts along uncemented joints clearances and creepage distances comply with 13.3 and 13.4                                                                                                                                   | EL 2008-10 |  |  |
|      | b) - Conductive parts along reliably cemented joints comply with 8.8                                                                                                                                                                        | EL 2008-11 |  |  |
|      | c) - Temperature cycle test ( 10 times):                                                                                                                                                                                                    | EL 2008-12 |  |  |
|      | d) - Dielectric strength test as per Cl.10.3                                                                                                                                                                                                | EL 2008-13 |  |  |
| 13.7 | a) - Enclosed, enveloped or hermetically sealed parts not conductively connected to the mains, clearances and creepage distances as in table 12<br>i) - Temperature cycle test (10 times),<br>ii) - Dielectric strength test as per Cl.10.3 | EL 2008-14 |  |  |
| 13.8 | Parts filled with insulating compound, meeting the requirements of 8.8                                                                                                                                                                      | EL 2008-15 |  |  |

Total number of requirements to be observed / inspected = 01

Total No. of Applicable Requirement =

No of requirement for which the sample passed =

Total number of tests to be conducted = 15

Total No. of Applicable Tests =

No. of tests for which the sample passed =

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

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(Approving Authority)

**Tests relating to Components****EL 2009- V1.0**

| Clause No | Test / Requirement name                                                                                                                                                               | Code       | Test result/ observation | Verdict |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------|---------|
| 14.       | Components*                                                                                                                                                                           | EL 2009-00 |                          |         |
| 14.1      | Flammability according to IEC 60695-11-10 or annex G, or 20.2.5                                                                                                                       | EL 2009-01 |                          |         |
| 14.2      | Resistors                                                                                                                                                                             |            |                          |         |
|           | Resistors separately approved                                                                                                                                                         | EL 2009-02 |                          |         |
|           | a) Resistors between hazardous live parts and accessible metal parts                                                                                                                  | EL 2009-03 |                          |         |
|           | b) Resistors, other than between hazardous live parts and accessible parts                                                                                                            | EL 2009-04 |                          |         |
| 14.3      | Capacitors and RC units                                                                                                                                                               | EL 2009-05 |                          |         |
|           | Capacitors separately approved :                                                                                                                                                      | EL 2009-06 |                          |         |
| 14.3.1    | Damp heat test duration 21 days                                                                                                                                                       | EL 2009-07 |                          |         |
| 14.3.2    | Y capacitors tested to IEC 60384-14:2005:                                                                                                                                             | EL 2009-08 |                          |         |
| 14.3.3    | X capacitors tested to IEC 60384-14:2005:                                                                                                                                             | EL 2009-09 |                          |         |
| 14.3.4    | Capacitors operating at mains frequency but not connected to the mains: tests for X2:                                                                                                 | EL 2009-10 |                          |         |
| 14.3.6    | Capacitors with volume exceeding 1750 mm <sup>3</sup> , where short-circuit current exceeds 0,2 A: compliance with IEC 60384-1, 4.38 category B or better :                           | EL 2009-11 |                          |         |
|           | Capacitors with volume exceeding 1750 mm <sup>3</sup> , mounted closer to a potential ignition source than table 13 permits: compliance with IEC 60384-1, 4.38 category B or better : | EL 2009-12 |                          |         |
| 14.4      | Inductors and windings                                                                                                                                                                | EL 2009-13 |                          |         |
| 14.4.1    | Comply with IEC 61558-1, IEC 61558-2                                                                                                                                                  | EL 2009-14 |                          |         |
|           | Addition to above insulating material complies with 20.2.5                                                                                                                            | EL 2009-15 |                          |         |
| 14.4.2    | Transformers and inductors marked with manufacturer's name and type:                                                                                                                  | EL 2009-16 |                          |         |
| 14.4.3    | General                                                                                                                                                                               |            |                          |         |
|           | Insulation material complies with clause 20.2.5                                                                                                                                       | EL 2009-17 |                          |         |
| 14.4.4    | Constructional requirements                                                                                                                                                           | EL 2009-18 |                          |         |

|          |                                                                                                                                                                                                     |            |  |  |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--|--|
| 14.4.4.1 | Clearances and creepage distances comply with clause 13                                                                                                                                             | EL 2009-19 |  |  |
| 14.4.4.2 | Transformers meet the constructional requirements                                                                                                                                                   | EL 2009-20 |  |  |
| 14.4.5   | Separation between windings                                                                                                                                                                         | EL 2009-21 |  |  |
| 14.4.5.1 | Class II transformers have adequate separation between hazardous live parts and accessible parts (double or reinforced insulation):                                                                 | EL 2009-22 |  |  |
|          | Coil formers and partition walls > 0,4 mm                                                                                                                                                           | EL 2009-23 |  |  |
| 14.4.5.2 | Class I transformers, with basic insulation and protective screening only if all 7 conditions are met                                                                                               | EL 2009-24 |  |  |
| 14.4.5.3 | Separating transformers with at least basic insulation                                                                                                                                              | EL 2009-25 |  |  |
| 14.4.6   | Insulation between hazardous live parts and accessible parts                                                                                                                                        | EL 2009-26 |  |  |
| 14.4.6.1 | Class II transformers have adequate insulation between hazardous live parts and accessible parts (double or reinforced insulation)                                                                  | EL 2009-27 |  |  |
|          | Coil formers and partition walls > 0,4 mm                                                                                                                                                           | EL 2009-28 |  |  |
| 14.4.6.2 | Class I transformers have adequate insulation between hazardous live parts and accessible conductive parts or those conductive parts or protective screens connected to a protective earth terminal | EL 2009-29 |  |  |
|          | Winding wires connected to protective earth have adequate current-carrying capacity                                                                                                                 | EL 2009-30 |  |  |
| 14.5     | High voltage components and assemblies (U > 4kV peak)                                                                                                                                               | EL 2009-31 |  |  |
| 14.5.1   | Component meets category V-1 of IEC 60695-11-10                                                                                                                                                     | EL 2009-32 |  |  |
| 14.5.2   | High voltage transformers and multipliers                                                                                                                                                           | EL 2009-33 |  |  |
| 14.5.3   | High voltage assemblies and other parts                                                                                                                                                             | EL 2009-34 |  |  |
| 14.6     | Protective devices                                                                                                                                                                                  | EL 2009-35 |  |  |
| 14.6.1   | Protective devices used within their ratings                                                                                                                                                        | EL 2009-36 |  |  |
|          | External clearances and creepage distances meet requirement of clause                                                                                                                               | EL 2009-37 |  |  |

|           |                                                                                                                                                                                                                 |            |  |  |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--|--|
|           | 13 for the voltage across the device when opened                                                                                                                                                                |            |  |  |
| 14.6.2    | Thermal Releases                                                                                                                                                                                                | EL 2009-38 |  |  |
| 14.6.2.1  | Comply with 14.6.2.2, 14.6.2.3 or 14.6.2.4                                                                                                                                                                      | EL 2009-39 |  |  |
| 14.6.2.2  | a) Thermal cut-outs separately approved                                                                                                                                                                         | EL 2009-40 |  |  |
|           | b) Thermal cut-outs tested as part of the submission                                                                                                                                                            | EL 2009-41 |  |  |
| 14.6.2.3  | a) Thermal links separately approved                                                                                                                                                                            | EL 2009-42 |  |  |
|           | b) Thermal links tested as part of the submission                                                                                                                                                               | EL 2009-43 |  |  |
| 14.6.2.4  | Thermal devices re-settable by soldering                                                                                                                                                                        | EL 2009-44 |  |  |
| 14.6.3    | Fuse-links and fuse holders*                                                                                                                                                                                    | EL 2009-45 |  |  |
| 14.6.3.1  | Fuse-links in the mains circuit according to IEC 60127                                                                                                                                                          | EL 2009-46 |  |  |
| 14.6.3.2  | Correct marking of fuse-links adjacent to holder :                                                                                                                                                              | EL 2009-47 |  |  |
| 14.6.3.3  | Not possible to connect fuses in parallel                                                                                                                                                                       | EL 2009-48 |  |  |
| 14.6.3.4  | Not possible to touch hazardous live parts when replacing fuse-links without the use of a tool :                                                                                                                | EL 2009-49 |  |  |
| 14.6.4    | PTC thermistors comply with IEC 60730-1:2010                                                                                                                                                                    | EL 2009-50 |  |  |
|           | PTC devices (>15 W) category V-1 or better                                                                                                                                                                      | EL 2009-51 |  |  |
| 14.6.5    | Circuit protectors have adequate breaking capacity and their position is correctly marked                                                                                                                       | EL 2009-52 |  |  |
| 14.7      | Switches*                                                                                                                                                                                                       | EL 2009-53 |  |  |
| 14.7.1 a) | Separate testing to IEC 61058-1 including:<br>- 10 000 operations<br>- Normal pollution suitability<br>- For CRT TV's, make and break speed independent of speed of actuation<br>- V-0 or compliance with G.1.1 | EL 2009-54 |  |  |
| 14.7.1 b) | Tested in the apparatus                                                                                                                                                                                         |            |  |  |
|           | Switch controlling > 0.2A with open contact voltage                                                                                                                                                             | EL 2009-55 |  |  |

|         |                                                                                                                                    |            |  |  |
|---------|------------------------------------------------------------------------------------------------------------------------------------|------------|--|--|
|         | > 35 V (peak) / 24 V dc complying with 14.6.3, 14.6.4 and V-0 or G.1.1                                                             |            |  |  |
|         | Switch controlling > 0.2A with open contact voltage<br>< 35 V (peak) / 24 V dc complying with 14.6.3 and V-0 or G.1.1              | EL 2009-56 |  |  |
|         | Switch controlling $\leq$ 0.2A with open contact voltage<br>> 35 V (peak)/24 V dc complying with 14.6.4 and V-0 or G.1.1           | EL 2009-57 |  |  |
| 14.7.2  | Switch tested to 14.7.1 b) checked according to IEC 61058-1 clause 13.1 and 10 000 operation test                                  | EL 2009-58 |  |  |
| 14.7.3  | Switch tested to 14.6.1 b) compliant with IEC 61058-1 subclause 16.2.2 d) and m) not attaining excessive temperatures in use       | EL 2009-59 |  |  |
| 14.7.4  | Switch tested to 14.6.1 b) has adequate dielectric strength                                                                        | EL 2009-60 |  |  |
| 14.7.5  | Mains switch controlling mains socket outlets additional tests to IEC 61058-1                                                      | EL 2009-61 |  |  |
| 14.8    | Safety interlocks according to 2.8 of IEC 60950-1                                                                                  | EL 2009-62 |  |  |
| 14.9    | Voltage setting device and the like are not likely to be changed accidentally                                                      | EL 2009-63 |  |  |
| 14.10   | Motors*                                                                                                                            | EL 2009-64 |  |  |
| 14.10.1 | a) Endurance test on motors                                                                                                        | EL 2009-65 |  |  |
|         | b) Motor start test Dielectric strength test                                                                                       | EL 2009-66 |  |  |
| 14.10.2 | Not adversely affected by oil or grease etc.                                                                                       | EL 2009-67 |  |  |
| 14.10.3 | Protection against moving parts                                                                                                    | EL 2009-68 |  |  |
| 14.10.4 | Motors with phase-shifting capacitors, three-phase motors and series motors meet clause. B.8, B.9 and B.10 of IEC 60950-1, Annex B | EL 2009-69 |  |  |
| 14.11   | Batteries*                                                                                                                         | EL 2009-70 |  |  |
| 14.11.1 | Comply with IEC 62133 if applicable *                                                                                              | EL 2009-71 |  |  |
|         | Batteries mounted with no risk of accumulation of flammable gases                                                                  | EL 2009-72 |  |  |
| 14.11.2 | No possibility of recharging user replaceable non-rechargeable batteries                                                           | EL 2009-73 |  |  |
| 14.11.3 | Recharging currents and times within                                                                                               | EL 2009-74 |  |  |

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|         |                                                                                                              |            |  |  |
|---------|--------------------------------------------------------------------------------------------------------------|------------|--|--|
|         | manufacturers limits                                                                                         |            |  |  |
|         | Lithium batteries discharge and reverse currents within the manufacturers limits                             | EL 2009-75 |  |  |
| 14.11.4 | Battery mould stress relief                                                                                  | EL 2009-76 |  |  |
| 14.11.5 | Battery drop test                                                                                            | EL 2009-77 |  |  |
| 14.12   | Opto couplers*                                                                                               | EL 2009-78 |  |  |
|         | Comply with constructional requirements of clause 8                                                          | EL 2009-79 |  |  |
|         | External clearances and creepage comply with 13.1                                                            | EL 2009-80 |  |  |
|         | Compound completely filling the casing or internal clearances and creepage comply with 13.1:                 | EL 2009-81 |  |  |
|         | a) Complies with 13.6 (jointed insulation) and N.3.2                                                         | EL 2009-82 |  |  |
|         | b) Complies with IEC 60747-5-5:2007                                                                          | EL 2009-83 |  |  |
|         | c) Complies with 13.8                                                                                        | EL 2009-84 |  |  |
| 14.13   | Surge suppression varistors*                                                                                 | EL 2009-85 |  |  |
|         | Comply with IEC 61051-2*                                                                                     | EL 2009-86 |  |  |
|         | Not connected between mains and accessible parts except for earthed parts of permanently connected apparatus | EL 2009-87 |  |  |
|         | GDT bridging basic insulation complies with electric strength and distance requirements                      | EL 2009-88 |  |  |
|         | Complies with the climatic, voltage, current pulse, fire hazard and thermal stress requirements of 14.13     | EL 2009-89 |  |  |

\*Total number of requirement to be observed / inspected = 09  
Total No. of Applicable Requirement =  
No of requirement for which the sample passed =

Total number of tests to be conducted = 81  
Total No. of Applicable Test =  
No. of tests for which the sample passed =

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

.....  
(Approving Authority)

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**Test s relating to wiring****EL 2010- V1.0**

| Clause No | Test / Requirement name                                                                                     | Code       | Test result/ observation | Verdict |
|-----------|-------------------------------------------------------------------------------------------------------------|------------|--------------------------|---------|
| 15        | Terminals *                                                                                                 | EL 2010-00 |                          |         |
| 15.1      | Plugs and Sockets*                                                                                          | EL 2010-01 |                          |         |
| 15.1.1    | Mains plug, appliance inlet, interconnection couplers and mains socket-outlet meet the appropriate standard | EL 2010-02 |                          |         |
|           | Overloading of plugs or appliance inlets prevented if the apparatus has mains socket outlets                | EL 2010-03 |                          |         |
|           | Overloading of internal wiring prevented if the apparatus has mains socket outlets                          | EL 2010-04 |                          |         |
| 15.1.2    | Design of connectors other than for mains power *                                                           | EL 2010-05 |                          |         |
|           | Design of sockets with symbol of 5.3 b) design *                                                            | EL 2010-06 |                          |         |
| 15.1.3    | Design of terminals and connectors used in output circuits of supply apparatus *                            | EL 2010-07 |                          |         |
| 15.2      | Provisions for protective earthing                                                                          | EL 2010-08 |                          |         |
|           | Accessible conductive parts of Class I equipment reliably connected to earth terminal, within equipment     | EL 2010-09 |                          |         |
|           | Protective earth conductors correctly fixed and coloured                                                    | EL 2010-10 |                          |         |
|           | Separate protective earth terminal near mains terminal and comply with 15.3                                 | EL 2010-11 |                          |         |
|           | Protective earth terminal resistant to corrosion                                                            | EL 2010-12 |                          |         |
|           | Earth resistance test: $< 0,1 \Omega$ at 25 A :                                                             | EL 2010-13 |                          |         |
| 15.3      | Terminals for external flexible cords and for permanent connection to the mains supply*                     | EL 2010-14 |                          |         |
| 15.3.1    | Adequate terminals for connection of permanent wiring*                                                      | EL 2010-15 |                          |         |
| 15.3.2    | Reliable connection of non-detachable cords                                                                 | EL 2010-16 |                          |         |
|           | Not soldered to conductors of a printed circuit board                                                       | EL 2010-17 |                          |         |
|           | Adequate clearances and creepage distances between connections should a wire break away                     | EL 2010-18 |                          |         |
|           | Wire secured by additional means to the conductor                                                           | EL 2010-19 |                          |         |
| 15.3.3    | Screws and nuts clamping conductors                                                                         | EL 2010-20 |                          |         |

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|        |                                                                                                                                         |            |  |  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------|------------|--|--|
|        | have adequate threads: ISO 261, ISO 262 or similar *                                                                                    |            |  |  |
| 15.3.4 | Conductors adequately fixed (two independent fixings) *                                                                                 | EL 2010-21 |  |  |
| 15.3.5 | Terminals allow connection of conductors having appropriate cross-sectional area                                                        | EL 2010-22 |  |  |
| 15.3.6 | Terminals to 15.3.3 have sizes required by table 16                                                                                     | EL 2010-23 |  |  |
| 15.3.7 | Terminals clamp conductors between metal and have adequate pressure                                                                     | EL 2010-24 |  |  |
|        | Terminals designed to avoid conductor slipping out when tightened                                                                       | EL 2010-25 |  |  |
|        | Terminals adequately fixed when tightened or loosened (no loosening, wiring not stressed, distances not reduced)                        | EL 2010-26 |  |  |
| 15.3.8 | Terminals carrying a current more than 0.2 A, contact pressure not transmitted by insulating material except ceramic*                   | EL 2010-27 |  |  |
| 15.3.9 | Termination of non-detachable cords: wires terminated near to each other                                                                | EL 2010-28 |  |  |
|        | Terminals located and shielded: test with 8 mm strand                                                                                   | EL 2010-29 |  |  |
| 15.4   | Devices forming a part of the mains plug*                                                                                               | EL 2010-30 |  |  |
| 15.4.1 | No undue strain on mains socket-outlets. Device shall be tested on the equipment as per Fig.11:<br>Torque to be applied : 0.25 Nm (Max) | EL 2010-31 |  |  |
| 15.4.2 | Device complies with standard for dimensions of mains plugs                                                                             | EL 2010-32 |  |  |
| 15.4.3 | Device has adequate mechanical strength (tests a,b,c)                                                                                   | EL 2010-33 |  |  |

\*Total number of requirements to be observed / inspected= 11

Total No. of Applicable Requirement: =

No of requirements for which the sample passed =

Total number of tests to be conducted = 23

Total No. of Applicable Tests: =

No. of tests for which the sample passed =

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

.....  
(Approving Authority)

**Test s relating to wiring****EL 2011- V1.0**

| Clause No | Test / Requirement name                                                                                                                                                                                                                                     | Code       | Test result/ observation | Verdict |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------|---------|
| 16        | External flexible cords*                                                                                                                                                                                                                                    | EL 2011-00 |                          |         |
| 16.1      | a) - Mains supply flexible cords shall be sheathed type, complying with IEC 60227 for PVC cords or as per IEC 60245 for synthetic rubber cords                                                                                                              | EL 2011-01 |                          |         |
|           | b) - Non-detachable cords for Class I have green/yellow core for protective earth                                                                                                                                                                           | EL 2011-02 |                          |         |
| 16.2      | Mains cords conductors shall have a nominal cross-sectional area not less than the values given in Table 18, for rated current consumption of the apparatus                                                                                                 | EL 2011-03 |                          |         |
| 16.3      | a) Flexible cords not complying with 16.1, used for interconnections between separate units of equipment used in combination and carrying hazardous live voltages, have adequate dielectric strength as per Cl.10.3                                         | EL 2011-04 |                          |         |
|           | b) Flexible cords not complying with 16.1, withstand bending and mechanical stress as per 3.1 of IEC 60227-2 read with except that Table-19 of this ISS applies<br>i) Flexing : 15000 times(30000 movements)<br>ii) Dielectric strength test as per Cl.10.3 | EL 2011-05 |                          |         |
| 16.4      | Flexible cords used for connection between equipment have adequate cross-sectional areas to avoid temperature rise under normal and fault conditions                                                                                                        | EL 2011-06 |                          |         |
| 16.5      | a) - Adequate strain relief on external flexible cords                                                                                                                                                                                                      | EL 2011-07 |                          |         |
|           | b) - Not possible to push cord back into equipment                                                                                                                                                                                                          | EL 2011-08 |                          |         |
|           | c) - Strain relief device unlikely to damage flexible cord                                                                                                                                                                                                  | EL 2011-09 |                          |         |
|           | d) - For mains cords of Class I equipment, hazardous live conductors become taut before earth conductor                                                                                                                                                     | EL 2011-10 |                          |         |
| 16.6      | Apertures for external flexible cord: no risk of damage to the cord during assembly or movement in use                                                                                                                                                      | EL 2011-11 |                          |         |

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|      |                                                                                                               |            |  |  |
|------|---------------------------------------------------------------------------------------------------------------|------------|--|--|
| 16.7 | a) Transportable apparatus fitted with detachable cord sets with appliance inlet as per IEC 60320-1 <b>or</b> | EL 2011-12 |  |  |
|      | b) Transportable apparatus shall have a means of stowage to protect the cord                                  | EL 2011-13 |  |  |

\*Total number of requirements to be observed / inspected = 01

Total No. of Applicable Requirement: =

No of requirements for which the sample passed =

Total number of tests to be conducted =13

Total No. of Applicable Tests =

No. of tests for which the sample passed =

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

.....  
(Approving Authority)

**Test s relating to wiring****EL 2012- V1.0**

| Clause No | Test / Requirement name                                                                                                                                                                      | Code       | Test result/ observation | Verdict |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------|---------|
| 17        | Electrical connections and mechanical fixings*                                                                                                                                               | EL 2012-00 |                          |         |
| 17.1      | a) - Screws are loosened and then tightened with a torque according to table 20                                                                                                              | EL 2012-01 |                          |         |
|           | b) -5 times in the case of screws operating in a thread of metal                                                                                                                             | EL 2012-02 |                          |         |
|           | c) -10 times in the case of screws operating in wood or in a thread in insulating material:                                                                                                  | EL 2012-03 |                          |         |
| 17.2      | Correct introduction of screws into female threads in non-metallic material                                                                                                                  | EL 2012-04 |                          |         |
| 17.3      | a) Screws or other fixing devices intended to fix Covers, legs, stands or the like, shall be captive in order to prevent replacement during servicing by screws or other fixing devices..... | EL 2012-05 |                          |         |
|           | b) Non-captive fixing screws: no hazard when replaced by a screw whose length is 10 times its diameter                                                                                       | EL 2012-06 |                          |         |
| 17.4      | No loosening of conductive parts carrying a current > 0.2 A                                                                                                                                  | EL 2012-07 |                          |         |
| 17.5      | Contact pressure not transmitted through plastic other than ceramic for connections carrying a current > 0.2 A *                                                                             | EL 2012-08 |                          |         |
| 17.6      | Stranded conductors of flexible supply cords carrying a current > 0.2 A with screw terminals not consolidated by solder *                                                                    | EL 2012-09 |                          |         |
| 17.7      | Cover fixing devices other than screws have adequate strength and their positioning is unambiguous                                                                                           | EL 2012-10 |                          |         |
| 17.8      | Detachable legs or stands supplied by the manufacturer of the apparatus shall be delivered with the relevant fixing means *                                                                  | EL 2012-11 |                          |         |
| 17.9      | a) - Internal pluggable connections, affecting safety, unlikely to become disconnected                                                                                                       | EL 2012-12 |                          |         |
|           | b) - By applying a 2N pull force in any direction to the connection, in case of doubt                                                                                                        | EL 2012-13 |                          |         |

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\*Total number of requirements to be observed / inspected = 04

Total No. of Applicable Requirement =

No of requirements for which the sample passed =

Total number of tests to be conducted = 10

Total No. of Applicable Tests =

No. of tests for which the sample passed =

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

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(Approving Authority)

**Tests relating to Physical Properties****EL 2013- V1.0**

| Clause No | Test / Requirement name                                                               | Code       | Test result/ observation | Verdict |
|-----------|---------------------------------------------------------------------------------------|------------|--------------------------|---------|
| 18        | MECHANICAL STRENGTH OF PICTURE TUBES AND PROTECTION AGAINST THE EFFECTS OF IMPLOSION* | EL 2013-00 |                          |         |
| 18.1      | Comply with IEC 61965 or 18.2                                                         | EL 2013-01 |                          |         |
| 18.2      | Non-intrinsically protected tubes                                                     | EL 2013-02 |                          |         |

Number of requirements to be observed / inspected = 01

Total No. of Applicable Requirement =

No of requirements for which the sample passed =

Number of tests to be conducted = 02

Total No. of Applicable Tests =

No. of tests to be conducted =

Certificate: It is certified that the above tests, were performed and found to be passing/failing in the requirement tested.

.....  
(Approving Authority)

**Tests relating to Physical Properties****EL 2014- V1.0**

| Clause No       | Test / Requirement name                                                                                                                     | Code       | Test result/ observation | Verdict |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------|---------|
| 19              | Stability and mechanical hazards*                                                                                                           | EL 2014-00 |                          |         |
| 19.1            | Apparatus > 7kg have adequate stability or is required to be fastened in place and provided with the warning of 5.5.2 f)                    | EL2014-01  |                          |         |
|                 | The test of 19.4<br>– apparatus with a mass of 25 kg or more, or                                                                            | EL2014-02  |                          |         |
|                 | – apparatus, excluding loudspeaker systems, with a height of 1 m or more, or                                                                | EL2014-03  |                          |         |
|                 | – apparatus, excluding loudspeaker systems, in combination with a supplied or recommended cart or stand with a total height of 1 m or more. | EL2014-04  |                          |         |
| 19.2            | Test at 10° to the horizontal                                                                                                               | EL2014-05  |                          |         |
| 19.3            | Vertical force test 100 N applied downwards                                                                                                 | EL2014-06  |                          |         |
| 19.4            | Horizontal force test, 100 N or 13% of weight, applied horizontally to point of least stability                                             | EL2014-07  |                          |         |
| 19.5            | Edges or corners not hazardous*                                                                                                             | EL2014-08  |                          |         |
| 19.6            | Mechanical strength of glass                                                                                                                | EL2014-09  |                          |         |
| 19.6.1          | Glass surfaces (exc.laminated) with an area exceeding 0,1 m <sup>2</sup> or major dimension > 450 mm, pass the test of 12.1.4               | EL2014-10  |                          |         |
| 19.6.2          | Fragmentation test                                                                                                                          | EL2014-11  |                          |         |
| 19.7            | Wall or ceiling mounting means                                                                                                              | EL2014-12  |                          |         |
| 19.7.1 - 19.7.3 | Not dislodged and remain mechanically intact after test according to 19.7.2 Test 1, Test 2 or Test 3:                                       | EL2014-13  |                          |         |

\*Total number of requirement to be tested = 02

Total No. of Applicable Requirement =

No of requirement for which the sample passed =

Total number of tests to be conducted = 12

Total No. of Applicable Tests =

No. of tests for which the sample passed =

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Certificate: It is certified that the above tests, were performed and found to be passing/failing in the requirement tested.

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(Approving Authority)

**Tests relating to Physical Properties****EL 2015- V1.0**

| Clause No | Test / Requirement name                                                                                                                                                                                                            | Code       | Test result/ observation | Verdict |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------|---------|
| 20        | Resistance to fire*                                                                                                                                                                                                                | EL2015-00  |                          |         |
| 20.1      | Start and spread of fire is prevented                                                                                                                                                                                              | EL 2015-01 |                          |         |
| 20.2      | Electrical components and mechanical parts                                                                                                                                                                                         | EL 2015-02 |                          |         |
| 20.2.1    | a) Exemption for components contained in an enclosure of material V-0 to IEC 60695-11-10 with openings not exceeding 1 mm in width                                                                                                 | EL 2015-03 |                          |         |
|           | b) Exemption for small components                                                                                                                                                                                                  | EL 2015-04 |                          |         |
| 20.2.2    | Electrical components meet the requirements of Clause 14 or 20.2.5                                                                                                                                                                 | EL 2015-05 |                          |         |
| 20.2.3    | Insulation of internal wiring working at voltages > 4 kV or leaving an internal fire enclosure, or located within the areas mentioned in Table 21, comply with G.2                                                                 | EL 2015-06 |                          |         |
| 20.2.4    | Material of printed circuit boards on which the available power exceeds 15 W at a voltage between 50 V and 400 V (peak) a.c. or d.c. meets V-1 or better to IEC 60695-11-10, unless used in a fire enclosure                       | EL 2015-07 |                          |         |
|           | Material of printed circuit boards on which the available power exceeds 15 W at a voltage >400 V (peak) a.c. or d.c. meets V-0 to IEC 60695-11-10.                                                                                 | EL 2015-08 |                          |         |
| 20.2.5    | Components and parts not covered by 20.1.1, 20.1.2 and 20.1.3 (other than fire enclosures) mounted nearer to a potential ignition source than the distances in Table 21 comply with the relevant flammability category in Table 21 | EL 2015-09 |                          |         |
|           | Components and parts as above but shielded from a potential ignition source, with the barrier area in accordance with Table 21 and fig. 13                                                                                         | EL 2015-10 |                          |         |
|           | Apparatus with voltages >4kV under normal operating conditions and distances to the enclosure exceed those specified Table 21, flammability classification HB40 or better is required for the enclosure                            | EL 2015-11 |                          |         |
| 20.3      | Fire enclosure*                                                                                                                                                                                                                    | EL 2015-12 |                          |         |

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|        |                                                                                                                      |            |  |  |
|--------|----------------------------------------------------------------------------------------------------------------------|------------|--|--|
| 20.3.1 | Potential ignition sources with open circuit voltage > 4 kV (peak) a.c. or d.c. contained in a fire enclosure to V-1 | EL 2015-13 |  |  |
| 20.3.2 | Internal fire enclosures with openings not exceeding 1 mm in width and with openings for wires completely filled     | EL 2015-14 |  |  |
| 20.3.3 | Requirements of 20.2.1 and 20.2.2 met by an internal fire enclosure                                                  | EL 2015-15 |  |  |

Total number of requirements to be tested = 02

Total No. of Applicable Requirement =

No of requirement for which the sample passed =

Total number of tests to be conducted = 14

Total No. of Applicable Tests =

No. of tests for which the sample passed =

Certificate: It is certified that the above tests, were performed and found to be passing/failing in the requirement tested.

.....  
(Approving Authority)

## Tests relating to Protection against Splashing Water

**EL 2016- V1.0**

| Clause No | Test / Requirement name                                                                | Code       | Test result/ observation | Verdict |
|-----------|----------------------------------------------------------------------------------------|------------|--------------------------|---------|
| ANNEX A   | Annex A, ADDITIONAL REQUIREMENTS FOR APPARATUS WITH PROTECTION AGAINST SPLASHING WATER | EL 2016-00 |                          |         |
| A.5       | Marking and instructions*                                                              | EL 2016-01 |                          |         |
| A.5.1     | A.5.2 i) Marked with at least IPX4 (IEC 60529)<br>5.5.2 a) does not apply              | EL 2016-02 |                          |         |
| A.10      | Insulation requirements*                                                               | EL 2016-03 |                          |         |
| A.10.3    | Splash and humidity treatment                                                          | EL 2016-04 |                          |         |
| A.10.3.1  | The enclosure provide adequate protection against splashing water                      | EL 2016-05 |                          |         |
| A.10.3.2  | Complies with 10.3, duration of the test is 168h                                       | EL 2016-06 |                          |         |

Number of requirements to be observed / inspected = 02

Total No. of Applicable Requirement =

No of requirements for which the sample passed =

Number of tests to be conducted = 05

Total No. of Applicable Tests =

No. of tests to be conducted =

Certificate: It is certified that the above tests, were performed and found to be passing/failing in the requirement tested.

.....  
(Approving Authority)

## Tests relating to Telecommunication Networks

## EL 2017- V1.0

| Clause No | Test / Requirement name                                           | Code       | Test result/ observation | Verdict |
|-----------|-------------------------------------------------------------------|------------|--------------------------|---------|
| ANNEX B   | Annex B, APPARATUS TO BE CONNECTED TO TELECOMMUNICATION NETWORKS* | EL 2017-00 |                          |         |
|           | Complies with IEC 62151 clause 1                                  | EL 2017-01 |                          |         |
|           | Complies with IEC 62151 clause 2                                  | EL 2017-02 |                          |         |
|           | Complies with IEC 62151 clause 3 modified                         | EL 2017-03 |                          |         |
|           | Complies with IEC 62151 clause 4 modified                         | EL 2017-04 |                          |         |
|           | Complies with IEC 62151 clause 5 modified                         | EL 2017-05 |                          |         |
|           | Complies with IEC 62151 clause 6                                  | EL 2017-06 |                          |         |
|           | Complies with IEC 62151 clause 7                                  | EL 2017-07 |                          |         |
|           | Complies with IEC 62151 annex A, B and C                          | EL 2017-08 |                          |         |

Number of requirements to be observed / inspected = 01

Total No. of Applicable Requirement =

No of requirements for which the sample passed =

Number of tests to be conducted = 08

Total No. of Applicable Tests =

No. of tests to be conducted =

Certificate: It is certified that the above tests, were performed and found to be passing/failing in the requirement tested.

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(Approving Authority)

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**Tests relating to Insulation Properties****EL 2018- V1.0**

| Cl. No. | Test / Requirement name                                | Test Code  | Test result/ observation | Verdict |
|---------|--------------------------------------------------------|------------|--------------------------|---------|
| Annex D | Annex D, MEASURING INSTRUMENTS FOR TOUCH-CURRENT TESTS | EL 2018-00 |                          |         |
| D.1     | Measuring instrument                                   | EL 2018-01 |                          |         |

\*- Total number of Requirements to be observed / inspected = 00

Total No of applicable Requirement =

No of Requirements for which the sample passed=

Total number of tests to be conducted = 02

Total No of applicable Tests =

No. of tests for which the sample passed=

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

(Approving Authority)

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**Tests relating to Electrical Safety****EL 2019-V1.0**

| Cl. No. | Test / Requirement name                                             | Test Code | Test result/ observation | Verdict |
|---------|---------------------------------------------------------------------|-----------|--------------------------|---------|
| Annex E | Annex E, MEASUREMENT OF CLEARANCES AND CREEPAGE DISTANCES (see 13 ) | EL2019-00 |                          |         |

\*- Total number of Requirements to be observed / inspected = 00

Total No of applicable Requirement =

No of Requirements for which the sample passed=

Total number of tests to be conducted = 01

Total No of applicable Tests =

No. of tests for which the sample passed=

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

(Approving Authority)

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**Tests relating to Electrical Safety****EL 2020-V1.0**

| Cl. No. | Test / Requirement name                      | Test Code  | Test result/ observation | Verdict |
|---------|----------------------------------------------|------------|--------------------------|---------|
| Annex F | Annex F, TABLE OF ELECTROCHEMICAL POTENTIALS | EL 2020-00 |                          |         |
|         | Metal(s) used :                              |            |                          |         |

\*- Total number of Requirements to be observed / inspected = 00

Total No of applicable Requirement =

No of Requirements for which the sample passed=

Total number of tests to be conducted = 01

Total No of applicable Tests =

No. of tests for which the sample passed=

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

.....  
(Approving Authority)

**Tests relating to Electrical Safety****EL 2021-V1.0**

| Cl. No. | Test / Requirement name                                  | Test Code | Test result/ observation | Verdict |
|---------|----------------------------------------------------------|-----------|--------------------------|---------|
| Annex G | Flammability test methods                                | EL2021-00 |                          |         |
| G.1     | Test as per IEC 60695-11-5                               | EL2021-01 |                          |         |
| G.1.1   | Flame test for V-0 material.                             | EL2021-02 |                          |         |
| G.1.2   | Flame test for V-1 material.                             | EL2021-03 |                          |         |
| G.1.3   | Flame test for V-2 material.                             | EL2021-04 |                          |         |
| G.1.4   | Flame test for HB75 or HB40 material.                    | EL2021-05 |                          |         |
| G.2     | Test for insulation of wires according to IEC 60695-11-5 | EL2021-06 |                          |         |
| G.3     | Test for barrier according to IEC 60695-11-5             | EL2021-07 |                          |         |

\*- Total number of Requirements to be observed / inspected = 00  
 Total No of applicable Requirement =

No of Requirements for which the sample passed=

Total number of tests to be conducted = 08  
 Total No of applicable Tests =

No. of tests for which the sample passed=

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

.....  
 (Approving Authority)

**Tests relating to Wiring****EL 2022-V1.0**

| Cl. No. | Test / Requirement name                                                           | Test Code | Test result/ observation | Verdict |
|---------|-----------------------------------------------------------------------------------|-----------|--------------------------|---------|
| Annex H | ANNEX H, INSULATED WINDING WIRES FOR USE WITHOUT INTERLEAVED INSULATION (SEE8.17) | EL2022-00 |                          |         |
| H.1     | GENERAL                                                                           | EL2022-01 |                          |         |
| H.2     | TYPE TESTS                                                                        | EL2022-02 |                          |         |
| H.2.1   | GENERAL                                                                           |           |                          |         |
| H.2.2   | ELECTRIC STRENGTH                                                                 | EL2022-03 |                          |         |
| H.2.3   | FLEXIBILITY AND ADHERENCE                                                         | EL2022-04 |                          |         |
| H.2.4   | HEAT SHOCK                                                                        | EL2022-05 |                          |         |
| H 2.5   | RETENTION OF ELECTRIC STRENGTH AFTER BENDING                                      | EL2022-06 |                          |         |
| H.3     | TESTING DURING MANUFACTURING                                                      | EL2022-07 |                          |         |
| H.3.1   | GENERAL                                                                           |           |                          |         |
| H.3.2   | ROUTINE TESTS                                                                     | EL2022-08 |                          |         |
| H 3.3   | SAMPLING TEST                                                                     | EL2022-09 |                          |         |

\*- Total number of Requirements to be observed / inspected = 00  
 Total No of applicable Requirement =

No of Requirements for which the sample passed=

Total number of tests to be conducted = 10  
 Total No of applicable Tests =

No. of tests for which the sample passed=

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

.....  
 (Approving Authority)

**Tests relating to Electrical safety****EL 2023-V1.0**

| Cl. No. | Test / Requirement name                                          | Test Code  | Test result/ observation | Verdict |
|---------|------------------------------------------------------------------|------------|--------------------------|---------|
| Annex J | Annex J, ALTERNATIVE METHOD FOR DETERMINING MINIMUM CLEARANCES   | EL 2023-00 |                          |         |
| J.1     | General                                                          |            |                          |         |
| J.2     | Summary of the procedure for determining minimum clearances      | EL 2023-01 |                          |         |
| J.3     | Determination of mains transient voltage (V)                     | EL 2023-02 |                          |         |
| J.4     | Determination of telecommunication network transient voltage (V) | EL 2023-03 |                          |         |
| J.5     | Determination of required withstand voltage (V)                  | EL 2023-04 |                          |         |
| J.5.1   | Mains transients and internal repetitive peaks                   | EL 2023-05 |                          |         |
| J.5.2   | Transients from telecommunication networks:                      | EL 2023-06 |                          |         |
| J.5.3   | Combination of transients                                        | EL 2023-07 |                          |         |
| J.5     | Measurement of transient voltages (V)                            | EL 2023-08 |                          |         |
|         | a) Transients from a mains supply                                |            |                          |         |
|         | For an a.c. mains supply                                         |            |                          |         |
|         | For a d.c. mains supply                                          |            |                          |         |
|         | b) Transients from a telecommunication network                   |            |                          |         |
| J.6     | Determination of minimum clearances                              | EL 2023-09 |                          |         |

\*- Total number of Requirements to be observed / inspected = 00

Total No of applicable Requirement =

No of Requirements for which the sample passed=

Total number of tests to be conducted..... = 10

Total No of applicable Tests =

No. of tests for which the sample passed=

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

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**Tests relating to Electrical safety****EL 2024-V1.0**

| Cl. No. | Test / Requirement name                                       | Test Code  | Test result/ observation | Verdict |
|---------|---------------------------------------------------------------|------------|--------------------------|---------|
| Annex K | Annex K, IMPULSE TEST GENERATORS (see 13.3.4 and annex J,J.5) | EL 2024-00 |                          |         |
| K1      | ITU-T impulse test generators                                 | EL 2024-01 |                          |         |

\*- Total number of Requirements to be observed / inspected = 00

Total No of applicable Requirement =

No of Requirements for which the sample passed=

Total number of tests to be conducted = 02

Total No of applicable Tests =

No. of tests for which the sample passed=

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

.....  
(Approving Authority)

**Tests relating to Electrical safety****EL 2025-V1.0**

| Clause No  | Test / Requirement name                                                                                                                 | Code       | Test result/ observation | Verdict |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------|---------|
| ANNEX L    | ANNEX L ,ADDITIONAL REQUIREMENTS FOR ELECTRONIC FLASH APPARATUS FOR PHOTOGRAPHIC PURPOSES*                                              | EL 2025-00 |                          |         |
| L.2        | General requirements                                                                                                                    |            |                          |         |
| L.4        | General test conditions                                                                                                                 |            |                          |         |
| L.5        | Marking and instructions*                                                                                                               | EL 2025-01 |                          |         |
| L.5.5.1    | Instructions for battery chargers and Supply apparatus indicating type or model number of flash apparatus with which it is to be used * | EL 2025-02 |                          |         |
|            | Instructions for flash apparatus indicating type or model number of battery chargers or Supply apparatus with which it is to be used *  | EL 2025-03 |                          |         |
| L.7        | Heating under normal operating conditions                                                                                               | EL 2025-04 |                          |         |
| L.7.1.6    | Lithium batteries meet permissible temp rise in Table 3 *                                                                               | EL 2025-05 |                          |         |
| L.9        | Electric shock hazard under normal operating conditions                                                                                 | EL 2025-06 |                          |         |
| L. 9.1.1.1 | Terminals for connection to synchroniser not hazardous live                                                                             | EL 2025-07 |                          |         |
| L.14       | Components                                                                                                                              | EL 2025-08 |                          |         |
| L.14.6.7   | Mains switch characteristics appropriate to its function under normal conditions                                                        | EL 2025-09 |                          |         |

\*- Total number of Requirements to be observed / inspected = 05

Total No of applicable Requirement =

No of Requirements for which the sample passed=

Total number of tests to be conducted = 05

Total No of applicable Tests =

No. of tests for which the sample passed=

Certificate: It is certified that the above tests were performed and found to be passing/failing in the requirement tested.

.....  
(Approving Authority)

**Table EL 2002-(00 to 06)**

|                               |                                        |    |                    |                    |                      |                      |                              |
|-------------------------------|----------------------------------------|----|--------------------|--------------------|----------------------|----------------------|------------------------------|
| <b>7.1</b>                    | <b>TABLE: Heating Test</b>             |    |                    |                    |                      |                      |                              |
|                               | <b>Ambient (°C) .....</b>              |    |                    |                    |                      |                      | —                            |
|                               | <b>Loudspeaker impedance (Ω) .....</b> |    |                    |                    |                      |                      | —                            |
| Cond.                         | U <sub>n</sub> (V)                     | Hz | I <sub>n</sub> (A) | P <sub>n</sub> (W) | U <sub>out</sub> (V) | P <sub>out</sub> (W) | Operating Condition / Status |
|                               |                                        |    |                    |                    |                      |                      |                              |
|                               |                                        |    |                    |                    |                      |                      |                              |
|                               |                                        |    |                    |                    |                      |                      |                              |
|                               |                                        |    |                    |                    |                      |                      |                              |
| <b>Test condition No.</b>     |                                        |    |                    | <b>No.</b>         | <b>No.</b>           | <b>No.</b>           | —                            |
| <b>Thermocouple Locations</b> |                                        |    |                    | <b>dT (K)</b>      | <b>dT (K)</b>        | <b>dT (K)</b>        | <b>dT (K) limit</b>          |
|                               |                                        |    |                    |                    |                      |                      |                              |
|                               |                                        |    |                    |                    |                      |                      |                              |
|                               |                                        |    |                    |                    |                      |                      |                              |
|                               |                                        |    |                    |                    |                      |                      |                              |
|                               |                                        |    |                    |                    |                      |                      |                              |
|                               |                                        |    |                    |                    |                      |                      |                              |
|                               |                                        |    |                    |                    |                      |                      |                              |
| Supplementary information:    |                                        |    |                    |                    |                      |                      |                              |

|                                    |                                               |                          |                          |               |                    |                         |
|------------------------------------|-----------------------------------------------|--------------------------|--------------------------|---------------|--------------------|-------------------------|
|                                    | <b>TABLE: Heating test, resistance method</b> |                          |                          |               |                    |                         |
|                                    | <b>Test condition No. ....</b>                |                          |                          |               |                    | —                       |
|                                    | <b>Ambient, t<sub>1</sub> (°C) .....</b>      |                          |                          |               |                    | —                       |
|                                    | <b>Ambient, t<sub>2</sub> (°C) .....</b>      |                          |                          |               |                    | —                       |
| <b>Temperature rise of winding</b> |                                               | <b>R<sub>1</sub> (Ω)</b> | <b>R<sub>2</sub> (Ω)</b> | <b>ΔT (K)</b> | <b>Max. dT (K)</b> | <b>Insulation class</b> |
|                                    |                                               |                          |                          |               |                    |                         |
|                                    |                                               |                          |                          |               |                    |                         |
|                                    |                                               |                          |                          |               |                    |                         |
| Supplementary information:         |                                               |                          |                          |               |                    |                         |

**Table EL 2002-07**

| <b>7.2</b>            | <b>TABLE: Heat Resistance of Insulating Materials</b> |                           |                      |
|-----------------------|-------------------------------------------------------|---------------------------|----------------------|
| Temperature T of part | T - normal conditions (°C)                            | T - fault conditions (°C) | Min T softening (°C) |
|                       |                                                       |                           |                      |
|                       |                                                       |                           |                      |
|                       |                                                       |                           |                      |
|                       |                                                       |                           |                      |

**Table EL 2005-(03 to 07)**

| <b>10.4</b>                                                  | <b>TABLE: Dielectric Strength</b> |                                |
|--------------------------------------------------------------|-----------------------------------|--------------------------------|
| Test voltage applied between:                                | Test potential applied (V)        | Breakdown / flashover (Yes/No) |
| Between mains poles (primary fuse disconnected)              |                                   |                                |
| Between parts separated by basic or supplementary insulation |                                   |                                |
| Between parts separated by double or reinforced insulation   |                                   |                                |
| Supplementary information:                                   |                                   |                                |

| <b>10.4</b>                                                  | <b>TABLE: Insulation Resistance Measurements</b> |                 |
|--------------------------------------------------------------|--------------------------------------------------|-----------------|
| Insulation resistance R between:                             | R (MΩ)                                           | Required R (MΩ) |
| Between mains poles (primary fuse disconnected)              |                                                  |                 |
| Between parts separated by basic or supplementary insulation |                                                  |                 |
| Between parts separated by double or reinforced insulation   |                                                  |                 |
| Supplementary information:                                   |                                                  |                 |

**Table EL 2006-(08 to 19)**

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|                            |           |                         |                    |                                             |  |
|----------------------------|-----------|-------------------------|--------------------|---------------------------------------------|--|
| 11                         |           | TABLE: Fault Conditions |                    |                                             |  |
| No.                        | Component | Fault                   | dT (K) / Component | Test conditions, test duration, test result |  |
|                            |           |                         |                    |                                             |  |
|                            |           |                         |                    |                                             |  |
|                            |           |                         |                    |                                             |  |
| Supplementary information: |           |                         |                    |                                             |  |

**Table EL 2008-(01 to 15)**

|                                                                |                                                     |                     |                       |                |          |                      |          |
|----------------------------------------------------------------|-----------------------------------------------------|---------------------|-----------------------|----------------|----------|----------------------|----------|
| 13                                                             | TABLE: Clearance And Creepage Distance Measurements |                     |                       |                |          |                      |          |
| Rated supply voltage..... :                                    |                                                     |                     | Pollution degree ...: |                |          | Material Group.....: |          |
| 2 N force on internal parts applied:                           |                                                     |                     |                       |                |          |                      |          |
| 30 N force on outside of conductive enclosure applied:         |                                                     |                     |                       |                |          |                      |          |
| clearance and creepage distance at/of:                         |                                                     | Working voltage (V) |                       | Clearance (mm) |          | Creepage (mm)        |          |
|                                                                |                                                     | U peak              | U r.m.s.              | Required       | Measured | required             | Measured |
| Primary (ac) to Earth (B)                                      |                                                     |                     |                       |                |          |                      |          |
| Primary (+dc) to Earth max (B)                                 |                                                     |                     |                       |                |          |                      |          |
| Primary (-dc) to Earth max (B)                                 |                                                     |                     |                       |                |          |                      |          |
| Across mains fuse F____ (B)                                    |                                                     |                     |                       |                |          |                      |          |
| Across primary directly connected to the mains (B)             |                                                     |                     |                       |                |          |                      |          |
| Hazardous live secondary to Earth (B)                          |                                                     |                     |                       |                |          |                      |          |
| Optocoupler input to output (R)                                |                                                     |                     |                       |                |          |                      |          |
| Primary to Secondary (R)                                       |                                                     |                     |                       |                |          |                      |          |
| Primary to accessible conductive parts (R)                     |                                                     |                     |                       |                |          |                      |          |
| Hazardous live secondary to non-hazardous live secondary (R)   |                                                     |                     |                       |                |          |                      |          |
| Hazardous live secondary to unearthed conductive enclosure (R) |                                                     |                     |                       |                |          |                      |          |
| Supplementary information:                                     |                                                     |                     |                       |                |          |                      |          |

**Photograph of marking plate and other photographs of the sample tested:**

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**\*\*End of Test Report\*\***